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Inventor(s)	Kottas; Gregg et al.

Heteroleptic iridium complexes as dopants

Abstract

Novel phosphorescent heteroleptic iridium complexes with phenylpyridine and dibenzo-containing ligands are provided. Alkyl substitution at specific positions on the ligands gives rise to compounds with improved OLED properties, including saturated green emission.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/746,351, filed May 17, 2022, which is a continuation of U.S. patent application Ser. No. 16/385,637, filed Apr. 16, 2019, now U.S. Pat. No. 11,380,854, which is a continuation of U.S. patent application Ser. No. 14/875,147, filed Oct. 5, 2015, now U.S. Pat. No. 10,312,458, which is a continuation of U.S. patent application Ser. No. 14/156,808 filed Jan. 16, 2014, now U.S. Pat. No. 9,184,397, which is a continuation of U.S. patent application Ser. No. 13/193,221 filed Jul. 28, 2011, now U.S. Pat. No. 8,709,615, and (2) U.S. patent application Ser. No. 16/385,637 is also a continuation-in-part of U.S. patent application Ser. No. 16/037,164, filed Jul. 7, 2018, now U.S. Pat. No. 11,637,251, which is a continuation of U.S. patent application Ser. No. 14/225,591, filed Mar. 26, 2014, now U.S. Pat. No. 10,056,566, which is a continuation of U.S. patent application Ser. No. 12/727,615, filed Mar. 19, 2010, now U.S. Pat. No. 8,722,205, which claims priority to U.S. Provisional Application No. 61/162,476, filed Mar. 23, 2009, the entire contents of which are incorporated herein by reference.

PARTIES TO A JOINT RESEARCH AGREEMENT

(1) The claimed invention was made by, on behalf of, and/or in connection with one or more of the following parties to a joint university corporation research agreement: Regents of the University of Michigan, Princeton University, The University of Southern California, and the Universal Display Corporation. The agreement was in effect on and before the date the claimed invention was made, and the claimed invention was made as a result of activities undertaken within the scope of the agreement.

FIELD OF THE INVENTION

(2) The present invention relates to heteroleptic iridium complexes suitable for inclusion in OLED devices.

BACKGROUND

(3) Opto-electronic devices that make use of organic materials are becoming increasingly desirable for a number of reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting devices (OLEDs), organic phototransistors, organic photovoltaic cells, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials. For example, the wavelength at which an organic emissive layer emits light may generally be readily tuned with appropriate dopants.

(4) OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as flat

panel displays, illumination, and backlighting. Several OLED materials and configurations are described in U.S. Pat. Nos. 5,844,363, 6,303,238, and 5,707,745, which are incorporated herein by reference in their entirety.

(5) One application for phosphorescent emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as “saturated” colors. In particular, these standards call for saturated red, green, and blue pixels. Color may be measured using CIE coordinates, which are well known to the art.

(6) One example of a green emissive molecule is tris(2-phenylpyridine) iridium, denoted Ir(ppy)₃, which has the following structure:

(7) ##STR00001##

(8) In this, and later figures herein, we depict the dative bond from nitrogen to metal (here, Ir) as a straight line.

(9) As used herein, the term “organic” includes polymeric materials as well as small molecule organic materials that may be used to fabricate organic opto-electronic devices. “Small molecule” refers to any organic material that is not a polymer, and “small molecules” may actually be quite large. Small molecules may include repeat units in some circumstances. For example, using a long chain alkyl group as a substituent does not remove a molecule from the “small molecule” class. Small molecules may also be incorporated into polymers, for example as a pendent group on a polymer backbone or as a part of the backbone. Small molecules may also serve as the core moiety of a dendrimer, which consists of a series of chemical shells built on the core moiety. The core moiety of a dendrimer may be a fluorescent or phosphorescent small molecule emitter. A dendrimer may be a “small molecule,” and it is believed that all dendrimers currently used in the field of OLEDs are small molecules.

(10) As used herein, “top” means furthest away from the substrate, while “bottom” means closest to the substrate. Where a first layer is described as “disposed over” a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is “in contact with” the second layer. For example, a cathode may be described as “disposed over” an anode, even though there are various organic layers in between.

(11) As used herein, “solution processible” means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

(12) A ligand may be referred to as “photoactive” when it is believed that the ligand directly contributes to the photoactive properties of an emissive material. A ligand may be referred to as “ancillary” when it is believed that the ligand does not contribute to the photoactive properties of an emissive material, although an ancillary ligand may alter the properties of a photoactive ligand.

(13) As used herein, and as would be generally understood by one skilled in the art, a first “Highest Occupied Molecular Orbital” (HOMO) or “Lowest Unoccupied Molecular Orbital” (LUMO) energy level is “greater than” or “higher than” a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A “higher” HOMO or LUMO energy level appears closer to the top of such a diagram than a “lower” HOMO or LUMO energy level.

(14) As used herein, and as would be generally understood by one skilled in the art, a first work function is “greater than” or “higher than” a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative to vacuum level, this means that a “higher” work function is more negative. On a conventional

energy level diagram, with the vacuum level at the top, a “higher” work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

(15) More details on OLEDs, and the definitions described above, can be found in U.S. Pat. No. 7,279,704, which is incorporated herein by reference in its entirety.

SUMMARY OF THE INVENTION

(16) In one aspect, a compound having the formula:

(17) ##STR00002##

is provided. R.sub.1 and R.sub.2 are optionally linked and the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2. R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked, and R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO.sub.2, CRR', SiRR', and GeRR', and R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and wherein n is 1 or 2.

(18) In one aspect, n is Z. In one aspect, X is O. In one aspect, R.sub.1 is hydrogen and R.sub.2 is alkyl. In another aspect, R.sub.1 is alkyl and R.sub.2 is hydrogen. In one aspect, R.sub.1 and R.sub.2 are alkyl. In one aspect, R.sub.1 and R.sub.2 contain one or more deuterium atoms. In another aspect, R.sub.1 or R.sub.2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl. In one aspect, R.sub.1 or R.sub.2 is iso-propyl.

(19) In one aspect, R.sub.1 or R.sub.2 contain one or more deuterium atoms. In one aspect, R.sub.3, R.sub.4, R.sub.5 and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof. In another aspect, at least one of R.sub.3, R.sub.4, R.sub.5 and R.sub.6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl. In one aspect, R.sub.3, R.sub.4, R.sub.5 or R.sub.6 contain one or more deuterium atoms.

(20) In one aspect, the compound is selected from the group consisting of Compound 53, Compounds 157-159, Compound 165, Compound 174, Compound 175, Compounds 184-185, Compound 314, Compound 321, Compounds 625-628, Compound 633, Compound 643, Compounds 652-653, and Compounds 1145-1146.

(21) In one aspect, a first device is provided. The first device comprises a first organic light-emitting device, further comprising an anode, a cathode, and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:

(22) ##STR00003##

R.sub.1 and R.sub.2 are optionally linked and the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2. R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked, and R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO.sub.2, CRR', SiRR', and GeRR', and R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and wherein n is 1 or 2.

(23) In one aspect, the first device is a consumer product. In another aspect, the first device is an organic light-emitting device. In another aspect, the first device comprises a lighting panel. In one aspect, the organic layer is an emissive layer and the compound is an emissive dopant. In another aspect, the organic layer is an emissive layer and the compound is a non-emissive dopant.

(24) In one aspect, the organic layer further comprises a host. In another aspect, the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan, wherein any substituent in the host is an unfused substituent independently selected from the group consisting of C.sub.nH.sub.2n+1, OC.sub.nH.sub.2n+1, Oar.sub.1, N(C.sub.nH.sub.2n+1).sub.2, N(Ar.sub.1)(Ar.sub.2), CH=CH—C.sub.nH.sub.2n+1, C≡CHC.sub.nH.sub.2n+1, Ar.sub.1, Ar.sub.1—Ar.sub.2, C.sub.nH.sub.2n—Ar.sub.1, or no substitution, wherein Ar.sub.1 and Ar.sub.2 are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof, and wherein n is from 1 to 10.

(25) In one aspect, the host has the formula

(26) ##STR00004##

(27) In another aspect, the host is selected from the group consisting of:

(28) ##STR00005## ##STR00006##

and combinations thereof.

(29) In one aspect, the host is a metal complex.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an organic light emitting device.

(2) FIG. 2 shows an inverted organic light-emitting device that does not have a separate electron transport layer.

(3) FIG. 3 shows a compound of Formula I.

DETAILED DESCRIPTION

(4) Generally, an OLED comprises at least one organic layer disposed between and electrically connected to an anode and a cathode. When a current is applied, the anode injects holes and the cathode injects electrons into the organic layer(s). The injected holes and electrons each migrate toward the oppositely charged electrode. When an electron and hole localize on the same molecule, an “exciton,” which is a localized electron-hole pair having an excited energy state, is formed. Light is emitted when the exciton relaxes via a photoemissive mechanism. In some cases, the exciton may be localized on an excimer or an exciplex. Non-radiative mechanisms, such as thermal relaxation, may also occur, but are generally considered undesirable.

(5) The initial OLEDs used emissive molecules that emitted light from their singlet states (“fluorescence”) as disclosed, for example, in U.S. Pat. No. 4,769,292, which is incorporated by reference in its entirety. Fluorescent emission generally occurs in a time frame of less than 10 nanoseconds.

(6) More recently, OLEDs having emissive materials that emit light from triplet states (“phosphorescence”) have been demonstrated. Baldo et al., “Highly Efficient Phosphorescent Emission from Organic Electroluminescent Devices,” Nature, vol. 395, 151-154, 1998; (“Baldo-I”) and Baldo et al., “Very high-efficiency green organic light-emitting devices based on electrophosphorescence,” Appl. Phys. Lett., vol. 75, No. 3, 4-6 (1999) (“Baldo-II”), which are incorporated by reference in their entireties. Phosphorescence is described in more detail in U.S. Pat. No. 7,279,704 at cols. 5-6, which are incorporated by reference.

(7) FIG. 1 shows an organic light emitting device **100**. The figures are not necessarily drawn to scale. Device **100** may include a substrate **110**, an anode **115**, a hole injection layer **120**, a hole transport layer **125**, an electron blocking layer **130**, an emissive layer **135**, a hole blocking layer **140**, an electron transport layer **145**, an electron injection layer **150**, a protective layer **155**, and a cathode **160**. Cathode **160** is a compound cathode having a first conductive layer **162** and a second conductive layer **164**. Device **100** may be fabricated by depositing the layers described, in order. The properties and functions of these various layers, as well as example materials, are described in

more detail in U.S. Pat. No. 7,279,704 at cols. 6-10, which are incorporated by reference.

(8) More examples for each of these layers are available. For example, a flexible and transparent substrate-anode combination is disclosed in U.S. Pat. No. 5,844,363, which is incorporated by reference in its entirety. An example of a p-doped hole transport layer is m-MTDATA doped with F.sub.4-TCNQ at a molar ratio of 50:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. Examples of emissive and host materials are disclosed in U.S. Pat. No. 6,303,238 to Thompson et al., which is incorporated by reference in its entirety. An example of an n-doped electron transport layer is Bphen doped with Li at a molar ratio of 1:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. U.S. Pat. Nos. 5,703,436 and 5,707,745, which are incorporated by reference in their entireties, disclose examples of cathodes including compound cathodes having a thin layer of metal such as Mg:Ag with an overlying transparent, electrically-conductive, sputter-deposited ITO layer. The theory and use of blocking layers is described in more detail in U.S. Pat. No. 6,097,147 and U.S. Patent Application Publication No. 2003/0230980, which are incorporated by reference in their entireties. Examples of injection layers are provided in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety. A description of protective layers may be found in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety.

(9) FIG. 2 shows an inverted OLED **200**. The device includes a substrate **210**, a cathode **215**, an emissive layer **220**, a hole transport layer **225**, and an anode **230**. Device **200** may be fabricated by depositing the layers described, in order. Because the most common OLED configuration has a cathode disposed over the anode, and device **200** has cathode **215** disposed under anode **230**, device **200** may be referred to as an “inverted” OLED. Materials similar to those described with respect to device **100** may be used in the corresponding layers of device **200**. FIG. 2 provides one example of how some layers may be omitted from the structure of device **100**.

(10) The simple layered structure illustrated in FIGS. 1 and 2 is provided by way of non-limiting example, and it is understood that embodiments of the invention may be used in connection with a wide variety of other structures. The specific materials and structures described are exemplary in nature, and other materials and structures may be used. Functional OLEDs may be achieved by combining the various layers described in different ways, or layers may be omitted entirely, based on design, performance, and cost factors. Other layers not specifically described may also be included. Materials other than those specifically described may be used. Although many of the examples provided herein describe various layers as comprising a single material, it is understood that combinations of materials, such as a mixture of host and dopant, or more generally a mixture, may be used. Also, the layers may have various sublayers. The names given to the various layers herein are not intended to be strictly limiting. For example, in device **200**, hole transport layer **225** transports holes and injects holes into emissive layer **220**, and may be described as a hole transport layer or a hole injection layer. In one embodiment, an OLED may be described as having an “organic layer” disposed between a cathode and an anode. This organic layer may comprise a single layer, or may further comprise multiple layers of different organic materials as described, for example, with respect to FIGS. 1 and 2.

(11) Structures and materials not specifically described may also be used, such as OLEDs comprised of polymeric materials (PLEDs) such as disclosed in U.S. Pat. No. 5,247,190 to Friend et al., which is incorporated by reference in its entirety. By way of further example, OLEDs having a single organic layer may be used. OLEDs may be stacked, for example as described in U.S. Pat. No. 5,707,745 to Forrest et al, which is incorporated by reference in its entirety. The OLED structure may deviate from the simple layered structure illustrated in FIGS. 1 and 2. For example, the substrate may include an angled reflective surface to improve out-coupling, such as a mesa structure as described in U.S. Pat. No. 6,091,195 to Forrest et al., and/or a pit structure as described in U.S. Pat. No. 5,834,893 to Bulovic et al., which are incorporated by reference in their entireties.

(12) Unless otherwise specified, any of the layers of the various embodiments may be deposited by any suitable method. For the organic layers, preferred methods include thermal evaporation, ink-jet, such as described in U.S. Pat. Nos. 6,013,982 and 6,087,196, which are incorporated by reference in their entireties, organic vapor phase deposition (OVPD), such as described in U.S. Pat. No. 6,337,102 to Forrest et al., which is incorporated by reference in its entirety, and deposition by organic vapor jet printing (OVJP), such as described in U.S. patent application Ser. No. 10/233,470, which is incorporated by reference in its entirety. Other suitable deposition methods include spin coating and other solution based processes. Solution based processes are preferably carried out in nitrogen or an inert atmosphere. For the other layers, preferred methods include thermal evaporation. Preferred patterning methods include deposition through a mask, cold welding such as described in U.S. Pat. Nos. 6,294,398 and 6,468,819, which are incorporated by reference in their entireties, and patterning associated with some of the deposition methods such as ink-jet and OVJD. Other methods may also be used. The materials to be deposited may be modified to make them compatible with a particular deposition method. For example, substituents such as alkyl and aryl groups, branched or unbranched, and preferably containing at least 3 carbons, may be used in small molecules to enhance their ability to undergo solution processing. Substituents having 20 carbons or more may be used, and 3-20 carbons is a preferred range. Materials with asymmetric structures may have better solution processability than those having symmetric structures, because asymmetric materials may have a lower tendency to recrystallize. Dendrimer substituents may be used to enhance the ability of small molecules to undergo solution processing.

(13) Devices fabricated in accordance with embodiments of the invention may be incorporated into a wide variety of consumer products, including flat panel displays, computer monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads up displays, fully transparent displays, flexible displays, laser printers, telephones, cell phones, personal digital assistants (PDAs), laptop computers, digital cameras, camcorders, viewfinders, micro-displays, vehicles, a large area wall, theater or stadium screen, or a sign. Various control mechanisms may be used to control devices fabricated in accordance with the present invention, including passive matrix and active matrix. Many of the devices are intended for use in a temperature range comfortable to humans, such as 18 degrees C. to 30 degrees C., and more preferably at room temperature (20-25 degrees C.).

(14) The materials and structures described herein may have applications in devices other than OLEDs. For example, other optoelectronic devices such as organic solar cells and organic photodetectors may employ the materials and structures. More generally, organic devices, such as organic transistors, may employ the materials and structures.

(15) The terms halo, halogen, alkyl, cycloalkyl, alkenyl, alkynyl, arylkyl, heterocyclic group, aryl, aromatic group, and heteroaryl are known to the art, and are defined in U.S. Pat. No. 7,279,704 at cols. 31-32, which are incorporated herein by reference.

(16) In one embodiment, a compound having the formula:

(17) ##STR00007##

is provided. R.sub.1 and R.sub.2 are optionally linked and the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2. Thus, both R.sub.1 and R.sub.2 both represent a substituent with at least one carbon. If R.sub.1 does not represent substituent containing carbon, then R.sub.2 must represent a substituent containing at least two carbons and vice versa. R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked, and R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO.sub.2, CRR', SiRR', and GeRR', and R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and wherein n is 1 or 2.

(18) In one embodiment, n is 2. In one embodiment, R.sub.1 is hydrogen and R.sub.2 is alkyl. In another embodiment, R.sub.1 is alkyl and R.sub.2 is hydrogen. In one embodiment, R.sub.1 and R.sub.2 are alkyl. In another embodiment, R.sub.1 or R.sub.2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl. In one embodiment, R.sub.1 or R.sub.2 is iso-propyl. Substitution at the 4- and 5-positions of either pyridine ring in the compounds of Formula I can give rise to compounds with desirable properties such as saturated green emission, high efficiencies, and long device lifetimes when incorporated in OLED devices. The photophysical and device properties of devices incorporating these compounds may be tuned by varying the nature of the substituent at the 4- or 5-position on the pyridine. The 4-position on a pyridine ring in the compound of Formula I is the position occupied by the R.sub.5 or R.sub.1 substituent, whereas the 5-position is the position occupied by the R.sub.4 or R.sub.2 substituent.

(19) As used herein, fragments containing the following structure:

(20) ##STR00008##

are called DBX groups, i.e. dibenzo X, where X is any of the atoms or groups described herein. Atoms A1-A8 can comprise nitrogen or carbon.

(21) In one embodiment, R.sub.1 or R.sub.2 contain one or more deuterium atoms. In one embodiment, R.sub.1 and R.sub.2 contain one or more deuterium atoms. In one embodiment, R.sub.3, R.sub.4, R.sub.5 and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof. In another embodiment, at least one of R.sub.3, R.sub.4, R.sub.5 and R.sub.6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl. In one embodiment, R.sub.3, R.sub.4, R.sub.5 or R.sub.6 contain one or more deuterium atoms. Without being bound by theory, incorporation of deuterium is thought to improve stability of compounds due the greater bond strength of the carbon-deuterium (C-D) bond versus the carbon-hydrogen (C—H) bond. Therefore, compounds wherein labile C—H bonds are replaced by C-D bonds, higher stability can be expected. Without being bound by theory, it is believed that incorporation of deuterium atoms on the alkyl groups of ligands for iridium complexes, the resulting complexes can have longer device lifetimes.

(22) In one embodiment, the compound is selected from the group consisting of:

(23) ##STR00009## ##STR00010## ##STR00011## ##STR00012## ##STR00013##
##STR00014##

(24) In one embodiment, suitable R.sub.1-R.sub.6 groups in the compounds of Formula I include the configurations of substituents in Table 1.

(25) TABLE-US-00001
TABLE 1
Compound # R.sub.1 R.sub.2 R.sub.3 R.sub.4 R.sub.5 R.sub.6
1. H Ethyl H H H H 2. H Ethyl Methyl H H H 3. H Ethyl H Methyl H H 4. H Ethyl H H Methyl H 5. H Ethyl H H H Methyl 6. H Ethyl Methyl Methyl H H 7. H Ethyl Methyl H Methyl H 8. H Ethyl Methyl H H Methyl 9. H Ethyl H Methyl Methyl H 10. H Ethyl H Methyl H Methyl 11. H Ethyl H H Methyl Methyl 12. H Ethyl Methyl Methyl Methyl H 13. H Ethyl Methyl Methyl H Methyl 14. H Ethyl Methyl H Methyl Methyl 15. H Ethyl H Methyl Methyl Methyl 16. H Ethyl Methyl Methyl Methyl Methyl 17. H Ethyl Ethyl H H H 18. H Ethyl H Ethyl H H 19. H Ethyl H H Ethyl H 20. H Ethyl H H H Ethyl 21. H Ethyl Methyl Ethyl H H 22. H Ethyl Methyl H Ethyl H 23. H Ethyl Methyl H H Ethyl 24. H Ethyl H Ethyl Methyl H 25. H Ethyl H Ethyl Ethyl H 26. H Ethyl H Methyl Ethyl H 27. H Ethyl Isopropyl H H H 28. H Ethyl H Isopropyl H H 29. H Ethyl H H Isopropyl H 30. H Ethyl H H H Isopropyl 31. H Ethyl Methyl Isopropyl H H 32. H Ethyl Methyl H Isopropyl H 33. H Ethyl Methyl H H Isopropyl 34. H Ethyl H Isopropyl Methyl H 35. H Ethyl H Isopropyl Isopropyl H 36. H Ethyl H Methyl Isopropyl H 37. H Ethyl H Isopropyl Ethyl H 38. H Ethyl H Ethyl Isopropyl H 39. H Ethyl Isobutyl H H H 40. H Ethyl H Isobutyl H H 41. H Ethyl H H Isobutyl H 42. H Ethyl H H H Isobutyl 43. H Ethyl Methyl Isobutyl H H 44. H Ethyl Methyl H Isobutyl H 45. H Ethyl Methyl H H Isobutyl 46. H Ethyl H Isobutyl Methyl H 47. H Ethyl H Isobutyl Isobutyl H 48. H Ethyl H Methyl Isobutyl H 49. H Ethyl H Isobutyl Ethyl H 50. H Ethyl H

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Methyl 1057. Isobutyl Isopropyl Ethyl H H H 1058. Isobutyl Isopropyl H Ethyl H H 1059. Isobutyl
Isopropyl H H Ethyl H 1060. Isobutyl Isopropyl H H H Ethyl 1061. Isobutyl Isopropyl Methyl
Ethyl H H 1062. Isobutyl Isopropyl Methyl H Ethyl H 1063. Isobutyl Isopropyl Methyl H H Ethyl
1064. Isobutyl Isopropyl H Ethyl Methyl H 1065. Isobutyl Isopropyl H Ethyl Ethyl H 1066.
Isobutyl Isopropyl H Methyl Ethyl H 1067. Isobutyl Isopropyl Isopropyl H H H 1068. Isobutyl
Isopropyl H Isopropyl H H 1069. Isobutyl Isopropyl H H Isopropyl H 1070. Isobutyl Isopropyl H
H H Isopropyl 1071. Isobutyl Isopropyl Methyl Isopropyl H H 1072. Isobutyl Isopropyl Methyl H
Isopropyl H 1073. Isobutyl Isopropyl Methyl H H Isopropyl 1074. Isobutyl Isopropyl H Isopropyl
Methyl H 1075. Isobutyl Isopropyl H Isopropyl Isopropyl H 1076. Isobutyl Isopropyl H Methyl
Isopropyl H 1077. Isobutyl Isopropyl H Isopropyl Ethyl H 1078. Isobutyl Isopropyl H Ethyl
Isopropyl H 1079. Isobutyl Isopropyl Isobutyl H H H 1080. Isobutyl Isopropyl H Isobutyl H H
1081. Isobutyl Isopropyl H H Isobutyl H 1082. Isobutyl Isopropyl H H H Isobutyl 1083. Isobutyl
Isopropyl Methyl Isobutyl H H 1084. Isobutyl Isopropyl Methyl H Isobutyl H 1085. Isobutyl
Isopropyl Methyl H H Isobutyl 1086. Isobutyl Isopropyl H Isobutyl Methyl H 1087. Isobutyl
Isopropyl H Isobutyl Isobutyl H 1088. Isobutyl Isopropyl H Methyl Isobutyl H 1089. Isobutyl
Isopropyl H Isobutyl Ethyl H 1090. Isobutyl Isopropyl H Ethyl Isobutyl H 1091. Isobutyl Isopropyl
H Isobutyl Isopropyl H 1092. Isobutyl Isopropyl H Isopropyl Isobutyl H 1093. Isobutyl Isobutyl H
H H H 1094. Isobutyl Isobutyl Methyl H H H 1095. Isobutyl Isobutyl H Methyl H H 1096. Isobutyl
Isobutyl H H Methyl H 1097. Isobutyl Isobutyl H H H Methyl 1098. Isobutyl Isobutyl Methyl
Methyl H H 1099. Isobutyl Isobutyl Methyl H Methyl H 1100. Isobutyl Isobutyl Methyl H H
Methyl 1101. Isobutyl Isobutyl H Methyl Methyl H 1102. Isobutyl Isobutyl H Methyl H Methyl
1103. Isobutyl Isobutyl H H Methyl Methyl 1104. Isobutyl Isobutyl Methyl Methyl Methyl H 1105.
Isobutyl Isobutyl Methyl Methyl H Methyl 1106. Isobutyl Isobutyl Methyl H Methyl Methyl 1107.
Isobutyl Isobutyl H Methyl Methyl Methyl 1108. Isobutyl Isobutyl Methyl Methyl Methyl Methyl
1109. Isobutyl Isobutyl Ethyl H H H 1110. Isobutyl Isobutyl H Ethyl H H 1111. Isobutyl Isobutyl H
H Ethyl H 1112. Isobutyl Isobutyl H H H Ethyl 1113. Isobutyl Isobutyl Methyl Ethyl H H 1114.
Isobutyl Isobutyl Methyl H Ethyl H 1115. Isobutyl Isobutyl Methyl H H Ethyl 1116. Isobutyl
Isobutyl H Ethyl Methyl H 1117. Isobutyl Isobutyl H Ethyl Ethyl H 1118. Isobutyl Isobutyl H
Methyl Ethyl H 1119. Isobutyl Isobutyl Isopropyl H H H 1120. Isobutyl Isobutyl H Isopropyl H H
1121. Isobutyl Isobutyl H H Isopropyl H 1122. Isobutyl Isobutyl H H H Isopropyl 1123. Isobutyl
Isobutyl Methyl Isopropyl H H 1124. Isobutyl Isobutyl Methyl H Isopropyl H 1125. Isobutyl
Isobutyl Methyl H H Isopropyl 1126. Isobutyl Isobutyl H Isopropyl Methyl H 1127. Isobutyl
Isobutyl H Isopropyl Isopropyl H 1128. Isobutyl Isobutyl H Methyl Isopropyl H 1129. Isobutyl
Isobutyl H Isopropyl Ethyl H 1130. Isobutyl Isobutyl H Ethyl Isopropyl H 1131. Isobutyl Isobutyl
Isobutyl H H H 1132. Isobutyl Isobutyl H Isobutyl H H 1133. Isobutyl Isobutyl H H Isobutyl H
1134. Isobutyl Isobutyl H H H Isobutyl 1135. Isobutyl Isobutyl Methyl Isobutyl H H 1136. Isobutyl
Isobutyl Methyl H Isobutyl H 1137. Isobutyl Isobutyl Methyl H H Isobutyl 1138. Isobutyl Isobutyl
H Isobutyl Methyl H 1139. Isobutyl Isobutyl H Isobutyl Isobutyl H 1140. Isobutyl Isobutyl H

Methyl Isobutyl H 1141. Isobutyl Isobutyl H Isobutyl Ethyl H 1142. Isobutyl Isobutyl H Ethyl Isobutyl H 1143. Isobutyl Isobutyl H Isobutyl Isopropyl H 1144. Isobutyl Isobutyl H Isopropyl Isobutyl H 1145. CD.sub.3CH.sub.2 H H H H H 1146. (CD.sub.3).sub.2CH H H H H H

(26) In one embodiment, a first device is provided. The first device comprises a first organic light emitting device, further comprising an anode, a cathode, and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:

(27) ##STR00015##

R.sub.1 and R.sub.2 are optionally linked and the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2. R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked, and R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, 502, CRR', SiRR', and GeRR', and R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and wherein n is 1 or 2.

(28) In one embodiment, the first device is a consumer product. In another embodiment, the first device is an organic light-emitting device. In another embodiment, the first device comprises a lighting panel. In one embodiment, the organic layer is an emissive layer and the compound is an emissive dopant. In another embodiment, the organic layer is an emissive layer and the compound is a non-emissive dopant.

(29) In one embodiment, the organic layer further comprises a host. In another embodiment, the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan, wherein any substituent in the host is an unfused substituent independently selected from the group consisting of C.sub.nH.sub.2n+1, OC.sub.nH.sub.2n+1, Oar.sub.1, N(C.sub.nH.sub.2n+1).sub.2, N(Ar.sub.1)(Ar.sub.2), CH=CH—C.sub.nH.sub.2n+1, C≡CHC.sub.nH.sub.2n+1, Ar.sub.1, Ar.sub.1—Ar.sub.2, C.sub.nH.sub.2n—Ar.sub.1, or no substitution, wherein Ar.sub.1 and Ar.sub.2 are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof, and wherein n is from 1 to 10.

(30) In one embodiment, the host has the formula I

(31) ##STR00016##

(32) In another embodiment, the host is selected from the group consisting of:

(33) ##STR00017## ##STR00018##

and combinations thereof.

(34) In one embodiment, the host is a metal complex.

Device Examples

(35) All example devices were fabricated by high vacuum (<10⁻⁷ Torr) thermal evaporation (VTE). The anode electrode is 1200 Å of indium tin oxide (ITO). The cathode consisted of 10 Å of LiF followed by 1,000 Å of Al. All devices are encapsulated with a glass lid sealed with an epoxy resin in a nitrogen glove box (<1 ppm of H.sub.2O and O.sub.2) immediately after fabrication, and a moisture getter was incorporated inside the package.

(36) The organic stack of the device examples consisted of sequentially, from the ITO surface, 100 Å of Compound C as the hole injection layer (HIL), 300 Å of 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (alpha-NPD) as the hole transporting layer (HTL), 300 Å of the invention compound doped in Compound D as host with 5-15 weight percent of a compound of Formula I as the emissive layer (EML), 50 Å of Compound D as blocking layer (BL), and 400 Å of Alq (tris-8-hydroxyquinoline aluminum) as the ETL. Comparative Examples with Compound A and Compound B were fabricated similarly to the Device Examples except that Compound A and Compound B were used as the emitter in the EML.

(37) The device results and data are summarized in Tables 1 from those devices. As used herein,

NPD, Alq, Compound A, Compound B, Compound C, and Compound D have the following structures:

(38) ##STR00019##

(39) TABLE-US-00002 TABLE 2 VTE Phosphorescent OLEDs Example HIL HTL EML (300 Å, doping %) BL ETL 1 Comparative Compound C NPD 300 Å Compound D Compound A Compound D Alq 400 Å Example 1 100 Å 10% 50 Å Comparative Compound C NPD 300 Å Compound D Compound B Compound D Alq 400 Å Example 2 100 Å 10% 50 Å Example 1 Compound C NPD 300 Å Compound D Compound 53 Compound D Alq 400 Å 100 Å 12% 50 Å Example 2 Compound C NPD 300 Å Compound D Compound 157 Compound D Alq 400 Å 100 Å 10% 50 Å Example 3 Compound C NPD 300 Å Compound D Compound 158 Compound D Alq 400 Å 100 Å 12% 50 Å Example 4 Compound C NPD 300 Å Compound D Compound 159 Compound D Alq 400 Å 100 Å 10% 50 Å Example 5 Compound C NPD 300 Å Compound D Compound 165 Compound D Alq 400 Å 100 Å 12% 50 Å Example 6 Compound C NPD 300 Å Compound D Compound 174 Compound D Alq 400 Å 100 Å 8% 50 Å Example 7 Compound C NPD 300 Å Compound D Compound 175 Compound D Alq 400 Å 100 Å 8% 50 Å Example 8 Compound C NPD 300 Å Compound D Compound 184 Compound D Alq 400 Å 100 Å 7% 50 Å Example 9 Compound C NPD 300 Å Compound D Compound 185 Compound D Alq 400 Å 100 Å 10% 50 Å Example 10 Compound C NPD 300 Å Compound D Compound 314 Compound D Alq 400 Å 100 Å 12% 50 Å Example 11 Compound C NPD 300 Å Compound D Compound 321 Compound D Alq 400 Å 100 Å 12% 50 Å Example 12 Compound C NPD 300 Å Compound D Compound 625 Compound D Alq 400 Å 100 Å 12% 50 Å Example 13 Compound C NPD 300 Å Compound D Compound 626 Compound D Alq 400 Å 100 Å 10% 50 Å Example 14 Compound C NPD 300 Å Compound D Compound 627 Compound D Alq 400 Å 100 Å 8% 50 Å Example 15 Compound C NPD 300 Å Compound D Compound 628 Compound D Alq 400 Å 100 Å 8% 50 Å Example 16 Compound C NPD 300 Å Compound D Compound 633 Compound D Alq 400 Å 100 Å 10% 50 Å Example 17 Compound C NPD 300 Å Compound D Compound 643 Compound D Alq 400 Å 100 Å 8% 50 Å Example 18 Compound C NPD 300 Å Compound D Compound 652 Compound D Alq 400 Å 100 Å 8% 50 Å Example 19 Compound C NPD 300 Å Compound D Compound 653 Compound D Alq 400 Å 100 Å 8% 50 Å Example 20 Compound C NPD 300 Å Compound D Compound 1145 Compound D Alq 400 Å 100 Å 10% 50 Å Example 21 Compound C NPD 300 Å Compound D Compound 1146 Compound D Alq 400 Å 100 Å 10% 50 Å

(40) TABLE-US-00003 TABLE 3 VTE Device Data $\lambda_{sub,max}$ FWHM Voltage LE EQE PE LT80% x y (nm) (nm) (V) (Cd/A) (%) (lm/W) (h) Comparative 0.367 0.607 532 66 6.5 43.2 11.5 21 250 Example 1 Compound A Comparative 0.336 0.623 524 66 5.2 58.2 15.7 35 170 Example 2 Compound B Example 1 0.345 0.619 526 64 5.1 69.2 18.7 42.7 20 Compound 53 Example 2 0.336 0.618 518 70 5.8 62.2 17.3 33.6 166 Compound 157 Example 3 0.318 0.632 518 60 4.7 73.4 20.2 49.1 55 Compound 158 Example 4 0.339 0.616 520 66 5.4 58.3 16.2 33.9 82 Compound 159 Example 5 0.333 0.626 522 62 4.6 56.1 15.2 38.3 54 Compound 165 Example 6 0.327 0.627 520 64 4.9 61.5 16.9 39.3 62 Compound 174 Example 7 0.314 0.635 518 60 5.3 71.2 19.5 42.2 77 Compound 175 Example 8 0.331 0.623 518 64 5.7 52.8 14.6 28.8 66 Compound 184 Example 9 0.331 0.623 518 64 5.4 56.7 15.7 32.9 53 Compound 185 Example 10 0.308 0.636 516 60 5.0 61.4 16.9 38.9 44 Compound 314 Example 11 0.326 0.630 520 60 5.1 61.4 16.6 37.7 73 Compound 321 Example 12 0.338 0.622 524 66 5.1 60.1 16.3 37.2 124 Compound 625 Example 13 0.336 0.624 522 64 5.5 67.4 18.2 38.2 71 Compound 626 Example 14 0.345 0.619 526 64 5.2 65.2 17.7 39.2 121 Compound 627 Example 15 0.351 0.615 526 64 5.6 68.6 18.6 38.7 124 Compound 628 Example 16 0.366 0.608 528 66 5.0 66.3 17.8 41.8 56 Compound 633 Example 17 0.339 0.625 526 60 5.2 71.5 19.1 43.3 108 Compound 643 Example 18 0.349 0.616 526 66 5.7 53.4 14.5 29.3 104 Compound 652 Example 19 0.337 0.626 524 62 5.1 65.6 17.6 40.8 84 Compound 653 Example 20 0.343 0.618 524 68 5.6 61.3 16.7 34.4 59 Compound 1145 Example 21 0.343 0.618 524 68 5.6 63 17.2 35.5 157 Compound 1146

(41) Table 3 is a summary of the device data. The luminous efficiency (LE), external quantum efficiency (EQE) and power efficiency (PE) were measured at 1000 nits, while the lifetime (LT.sub.80%) was defined as the time required for the device to decay to 80% of its initial luminance under a constant current density of 40 mA/cm.sup.2.

(42) The advantage of alkyl substitution on the 4- and 5-positions of the DBX-pyridine ring is clear from Table 3. Compared to Comparative Example 1, with no substitution at the 4- or 5-positions of the DBX-pyridine ring, the compounds of Formula I are more saturated (lower CIE x coordinate and shorter $\lambda_{\text{sub.max}}$) and the broadness as measured by the FWHM is comparable. In all the inventive compounds, the voltage is lower and the LE, PE and EQE values are all higher. In the case of Compounds 53, 158, 175, 633 and 643, the PE is at least twice as high as Comparative Example 1.

(43) Compared to Comparative Example 2 (Compound B), with only one-carbon substitution (methyl) at the 4-position of the DBX-pyridine ring, Compounds 53, 158, 174, 175, 184, 185, and 314 have more saturated color based CIE x coordinate and Compounds 157, 158, 159, 165, 174, 175, 184, 185, 314, 321 and 626 all have shorter $\lambda_{\text{sub.max}}$ values. Most of the compounds of Formula I have narrower emission profiles (as measured by FWHM) than Comparative Example 2. Compounds 53, 158, 165, 314, 321, 625, 633 and 653 all have lower driving voltages than Comparative Example 2. Most of the compounds of Formula I have greater LE, PE and EQE values than Comparative Example 2.

(44) Combination with Other Materials

(45) The materials described herein as useful for a particular layer in an organic light emitting device may be used in combination with a wide variety of other materials present in the device. For example, emissive dopants disclosed herein may be used in conjunction with a wide variety of hosts, transport layers, blocking layers, injection layers, electrodes and other layers that may be present. The materials described or referred to below are non-limiting examples of materials that may be useful in combination with the compounds disclosed herein, and one of skill in the art can readily consult the literature to identify other materials that may be useful in combination.

(46) HIL/HTL:

(47) A hole injecting/transporting material to be used in the present invention is not particularly limited, and any compound may be used as long as the compound is typically used as a hole injecting/transporting material. Examples of the material include, but not limit to: a phthalocyanine or porphyrin derivative; an aromatic amine derivative; an indolocarbazole derivative; a polymer containing fluorohydrocarbon; a polymer with conductivity dopants; a conducting polymer, such as PEDOT/PSS; a self-assembly monomer derived from compounds such as phosphonic acid and silane derivatives; a metal oxide derivative, such as MoO.sub.x; a p-type semiconducting organic compound, such as 1,4,5,8,9,12-Hexaazatriphenylenehexacarbonitrile; a metal complex, and a cross-linkable compounds.

(48) Examples of aromatic amine derivatives used in HIL or HTL include, but not limit to the following general structures:

(49) ##STR00020##

(50) Each of Ar.sup.1 to Ar.sup.9 is selected from the group consisting aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, azulene; group consisting aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine,

benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and group consisting 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Wherein each Ar is further substituted by a substituent selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

(51) In one aspect, Ar.sup.1 to Ar.sup.9 is independently selected from the group consisting of:

(52) ##STR00021##

k is an integer from 1 to 20; X.sup.1 to X.sup.8 is C (including CH) or N; Ar.sup.1 has the same group defined above.

(53) Examples of metal complexes used in HIL or HTL include, but not limit to the following general formula:

(54) ##STR00022## M is a metal, having an atomic weight greater than 40; (Y.sup.1—Y.sup.2) is a bidentate ligand, Y.sup.1 and Y.sup.2 are independently selected from C, N, O, P, and S; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and m+n is the maximum number of ligands that may be attached to the metal.

(55) In one aspect, (Y.sup.1—Y.sup.2) is a 2-phenylpyridine derivative.

(56) In another aspect, (Y.sup.1—Y.sup.2) is a carbene ligand.

(57) In another aspect, M is selected from Ir, Pt, Os, and Zn.

(58) In a further aspect, the metal complex has a smallest oxidation potential in solution vs. Fc.sup.+ / Fc couple less than about 0.6 V.

(59) Host:

(60) The light emitting layer of the organic EL device of the present invention preferably contains at least a metal complex as light emitting material, and may contain a host material using the metal complex as a dopant material. Examples of the host material are not particularly limited, and any metal complexes or organic compounds may be used as long as the triplet energy of the host is larger than that of the dopant.

(61) Examples of metal complexes used as host are preferred to have the following general formula.

(62) ##STR00023## M is a metal; (Y.sup.3—Y.sup.4) is a bidentate ligand, Y.sup.3 and Y.sup.4 are independently selected from C, N, O, P, and S; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and m+n is the maximum number of ligands that may be attached to the metal.

(63) In one aspect, the metal complexes are:

(64) ##STR00024## (O—N) is a bidentate ligand, having metal coordinated to atoms O and N.

(65) In another aspect, M is selected from Ir and Pt.

(66) In a further aspect, (Y.sup.3—Y.sup.4) is a carbene ligand.

(67) Examples of organic compounds used as host are selected from the group consisting aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, azulene; group consisting aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole,

benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyrindine, benzothienopyridine, thienodipyrindine, benzoselenophenopyridine, and selenophenodipyrindine; and group consisting 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Wherein each group is further substituted by a substituent selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

(68) In one aspect, host compound contains at least one of the following groups in the molecule:

(69) ##STR00025## R^{sup.1} to R^{sup.7} is independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above. k is an integer from 0 to 20. X^{sup.1} to X^{sup.8} is selected from C (including CH) or N.

HBL:

(70) A hole blocking layer (HBL) may be used to reduce the number of holes and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED.

(71) In one aspect, compound used in HBL contains the same molecule used as host described above.

(72) In another aspect, compound used in HBL contains at least one of the following groups in the molecule:

(73) ##STR00026## k is an integer from 0 to 20; L is an ancillary ligand, m is an integer from 1 to 3.

ETL:

(74) Electron transport layer (ETL) may include a material capable of transporting electrons. Electron transport layer may be intrinsic (undoped), or doped. Doping may be used to enhance conductivity. Examples of the ETL material are not particularly limited, and any metal complexes or organic compounds may be used as long as they are typically used to transport electrons.

(75) In one aspect, compound used in ETL contains at least one of the following groups in the molecule:

(76) ##STR00027## R^{sup.1} is selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, it has the similar definition as Ar's mentioned above. Ar^{sup.1} to Ar^{sup.3} has the similar definition as Ar's mentioned above. k is an integer from 0 to 20. X^{sup.1} to X^{sup.8} is selected from C (including CH) or N.

(77) In another aspect, the metal complexes used in ETL contains, but not limit to the following general formula:

(78) ##STR00028##

(79) (O—N) or (N—N) is a bidentate ligand, having metal coordinated to atoms O, N or N, N; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal.

(80) In any above-mentioned compounds used in each layer of the OLED device, the hydrogen

atoms can be partially or fully deuterated.

(81) In addition to and/or in combination with the materials disclosed herein, many hole injection materials, hole transporting materials, host materials, dopant materials, exciton/hole blocking layer materials, electron transporting and electron injecting materials may be used in an OLED. Non-limiting examples of the materials that may be used in an OLED in combination with materials disclosed herein are listed in Table 3 below. Table 3 lists non-limiting classes of materials, non-limiting examples of compounds for each class, and references that disclose the materials.

(82) TABLE-US-00004 TABLE 2 MATERIAL EXAMPLES OF MATERIAL PUBLICATIONS

Hole injection materials Phthalocyanine and porphyrin compounds  embedded image Appl. Phys. Lett. 69, 2160 (1996) Starburst triarylamine  embedded image J. Lumin. 72-74, 985 (1997) CF.sub.x Fluorohydrocarbon polymer  embedded image Appl. Phys. Lett. 78, 673 (2001) Conducting polymers (e.g., PEDOT:PSS, polyaniline, polythiophene)  embedded image Synth. Met. 87, 171 (1997) WO2007002683 Phosphonic acid and silane SAMs  embedded image US20030162053 Triarylamine or polythiophene polymers with conductivity dopants  embedded image EA01725079A1  embedded image  embedded image Arylamines complexed with metal oxides such as molybdenum and tungsten oxides  embedded image SID Symposium Digest, 37, 923 (2006) WO2009018009 p-type semiconducting organic complexes  embedded image US20020158242 Metal organometallic complexes  embedded image US20060240279 Cross-linkable compounds  embedded image US20080220265 Hole transporting materials Triarylamine (e.g., TPD, α -NPD)  embedded image Appl. Phys. Lett. 51, 913 (1987)  embedded image U.S. Pat. No. 5,061,569  embedded image EP650955  embedded image J. Mater. Chem. 3, 319 (1993)  embedded image Appl. Phys. Lett. 90, 183503 (2007)  embedded image Appl. Phys. Lett. 90, 183503 (2007) Triarylamine on spirofluorene core  embedded image Synth. Met. 91, 209 (1997) Arylamine carbazole compounds  embedded image Adv. Mater. 6, 677 (1994), US20080124572 Triarylamine with (di)benzothiophene/ (di)benzofuran  embedded image US20070278938, US20080106190 Indolocarbazoles  embedded image Synth. Met. 111, 421 (2000) Isoindole compounds  embedded image Chem. Mater. 15, 3148 (2003) Metal carbene complexes  embedded image US20080018221 Phosphorescent OLED host materials Red hosts Arylcarbazoles  embedded image Appl. Phys. Lett. 78, 1622 (2001) Metal 8-hydroxyquinolates (e.g., Alq.sub.3, BALq)  embedded image Nature 395, 151 (1998)  embedded image US20060202194  embedded image WO2005014551  embedded image WO2006072002 Metal phenoxybenzothiazole compounds  embedded image Appl. Phys. Lett. 90, 123509 (2007) Conjugated oligomers and polymers (e.g., polyfluorene)  embedded image Org. Electron. 1, 15 (2000) Aromatic fused rings  embedded image WO2009066779, WO2009066778, WO2009063833, US20090045731, US20090045730, WO2009008311, US20090008605, US20090009065 Zinc complexes  embedded image WO2009062578 Green hosts Arylcarbazoles  embedded image Appl. Phys. Lett. 78, 1622 (2001)  embedded image US20030175553  embedded image WO2001039234 Aryltriphenylene compounds  embedded image US20060280965  embedded image US20060280965  embedded image WO2009021126 Donor acceptor type molecules  embedded image WO2008056746 Aza-carbazole/DBT/DBF  embedded image JP2008074939 Polymers (e.g., PVK)  embedded image Appl. Phys. Lett. 77, 2280 (2000) Spirofluorene compounds  embedded image WO2004093207 Metal phenoxybenzoxazole compounds  embedded image WO2005089025  embedded image WO2006132173  embedded image JP200511610 Spirofluorene-carbazole compounds  embedded image JP2007254297  embedded image JP2007254297 Indolocarbazoles  embedded image WO2007063796  embedded image WO2007063754 5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole)  embedded image J. Appl. Phys. 90, 5048 (2001)  embedded image WO2004107822 Tetraphenylene complexes  embedded image US20050112407 Metal phenoxypyridine compounds  embedded image WO2005030900 Metal

coordination complexes (e.g., Zn, Al with N{circumflex over ()}N ligands)  embedded image
US20040137268, US20040137267 Blue hosts Arylcarbazoles  embedded image Appl. Phys. Lett.
82, 2422 (2003)  embedded image US20070190359 Dibenzothiophene/ Dibenzofuran-carbazole
compounds  embedded image WO2006114966, US20090167162  embedded image
US20090167162  embedded image WO2009086028  embedded image US20090030202,
US20090017330 Silicon aryl compounds 0  embedded image US20050238919  embedded image
WO2009003898 Silicon/Germanium aryl compounds  embedded image EP2034538A Aryl
benzoyl ester  embedded image WO2006100298 High triplet metal organometallic complex
 embedded image U.S. Pat. No. 7,154,114 Phosphorescent dopants Red dopants Heavy metal
porphyrins (e.g., PtOEP)  embedded image Nature 395, 151 (1998) Iridium(III) organometallic
complexes  embedded image Appl. Phys. Lett. 78, 1622 (2001)  embedded image
US2006835469  embedded image US2006835469  embedded image US20060202194 00
 embedded image US20060202194 01  embedded image US20070087321 02  embedded image
US20070087321 03  embedded image Adv. Mater. 19, 739 (2007) 04  embedded image
WO2009100991 05  embedded image WO2008101842 Platinum(II) organometallic complexes 06
 embedded image WO2003040257 Osmium(III) complexes 07  embedded image Chem. Mater.
17, 3532 (2005) Ruthenium(II) complexes 08  embedded image Adv. Mater. 17, 1059 (2005)
Rhenium (I), (II), and (III) complexes 09  embedded image US20050244673 Green dopants
Iridium(III) organometallic complexes 0  embedded image and its derivatives Inorg. Chem. 40,
1704 (2001)  embedded image US20020034656  embedded image U.S. Pat. No. 7,332,232
 embedded image US20090108737  embedded image US20090039776  embedded image U.S.
Pat. No. 6,921,915  embedded image U.S. Pat. No. 6,687,266  embedded image Chem. Mater.
16, 2480 (2004)  embedded image US20070190359  embedded image US 20060008670
JP2007123392 0  embedded image Adv. Mater. 16, 2003 (2004)  embedded image Angew.
Chem. Int. Ed. 2006, 45, 7800  embedded image WO2009050290  embedded image
US20090165846  embedded image US20080015355 Monomer for polymeric metal
organometallic compounds  embedded image U.S. Pat. No. 7,250,226, U.S. Pat. No. 7,396,598
Pt(II) organometallic complexes, including polydentated ligands  embedded image Appl. Phys.
Lett. 86, 153505 (2005)  embedded image Appl. Phys. Lett. 86, 153505 (2005)
 embedded image Chem. Lett. 34, 592 (2005)  embedded image WO2002015645 0
 embedded image US20060263635 Cu complexes  embedded image WO2009000673 Gold
complexes  embedded image Chem. Commun. 2906 (2005) Rhenium(III) complexes
 embedded image Inorg. Chem. 42, 1248 (2003) Deuterated organometallic complexes
 embedded image US20030138657 Organometallic complexes with two or more metal centers
 embedded image US20030152802  embedded image U.S. Pat. No. 7,090,928 Blue dopants
Iridium(III) organometallic complexes  embedded image WO2002002714  embedded image
WO2006009024  embedded image US20060251923 0  embedded image U.S. Pat. No.
7,393,599, WO2006056418, US20050260441, WO2005019373  embedded image U.S. Pat. No.
7,534,505  embedded image U.S. Pat. No. 7,445,855  embedded image US20070190359,
US20080297033  embedded image U.S. Pat. No. 7,338,722  embedded image US20020134984
 embedded image Angew. Chem. Int. Ed. 47, 1 (2008)  embedded image Chem. Mater. 18, 5119
(2006)  embedded image Inorg. Chem. 46, 4308 (2007)  embedded image WO2005123873 0
 embedded image WO2005123873 embedded image WO2007004380 embedded image
WO2006082742 Osmium(II) complexes embedded image U.S. Pat. No. 7,279,704
 embedded image Organometallics 23, 3745 (2004) Gold complexes embedded image Appl.
Phys. Lett. 74, 1361 (1999) Platinum(II) complexes embedded image WO2006098120,
WO2006103874 Exciton/hole blocking layer materials Bathocuproine compounds (e.g., BCP,
BPhen) embedded image Appl. Phys. Lett. 75, 4 (1999) embedded image Appl. Phys. Lett. 79,
449 (2001) Metal 8-hydroxyquinolates (e.g., BA1q) embedded image Appl. Phys. Lett. 81, 162
(2002) 5-member ring electron deficient heterocycles such as triazole, oxadiazole, imidazole,

benzimidazole 0  Appl. Phys. Lett. 81, 162 (2002) Triphenylene compounds  US20050025993 Fluorinated aromatic compounds  Appl. Phys. Lett. 79, 156 (2001) Phenothiazine-S-oxide  WO2008132085 Electron transporting materials Anthracene- benzimidazole compounds  WO2003060956  US20090179554 Aza triphenylene derivatives  US20090115316 Anthracene-benzothiazole compounds  Appl. Phys. Lett. 89, 063504 (2006) Metal 8-hydroxyquinolates (e.g., Alq.sub.3, Zrq.sub.4)  Appl. Phys. Lett. 51, 913 (1987) U.S. Pat. No. 7,230,107 Metal hydroxybenzoquinolates  Chem. Lett. 5, 905 (1993) Bathocuprine compounds such as BCP, BPhen, etc 0  Appl. Phys. Lett. 91, 263503 (2007)  Appl. Phys. Lett. 79, 449 (2001) 5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole, imidazole, benzimidazole)  Appl. Phys. Lett. 74, 865 (1999)  Appl. Phys. Lett. 55, 1489 (1989)  Jpn. J. Apply. Phys. 32, L917 (1993) Silole compounds  Org. Electron. 4, 113 (2003) Arylborane compounds  J. Am. Chem. Soc. 120, 9714 (1998) Fluorinated aromatic compounds  J. Am. Chem. Soc. 122, 1832 (2000) Fullerene (e.g., C60)  US20090101870 Triazine complexes  US20040036077 Zn (N{circumflex over ()}N) complexes 0  U.S. Pat. No. 6,528,187

EXPERIMENTAL

(83) Chemical abbreviations used throughout this document are as follows: Cy is cyclohexyl, dba is dibenzylideneacetone, EtOAc is ethyl acetate, DME is dimethoxyethane, dppe is 1,2-bis(diphenylphosphino)ethane, THE is tetrahydrofuran, DCM is dichloromethane, S-Phos is dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine.

(84) ##STR00181##

(85) Synthesis of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine: Dibenzo[b,d]furan-4-ylboronic acid (9.5 g, 44.8 mmol), 2,5-dichloropyridine (7.0 g, 47.0 mmol), Pd(PPh.sub.3).sub.4 (2.6 g, 2.2 mmol) and potassium carbonate (18.6 g, 134 mmol) were added to dimethoxyethane (75 mL) and water (75 mL). The reaction mixture was degassed with nitrogen before being heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane to yield 11.7 g of crude product. The product was crystallized from hexane to give 9.5 g (76%) of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine as white needles. The product was confirmed by GC/MS.

(86) ##STR00182##

(87) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine: 5-Chloro-2-(dibenzo[b,d]furan-4-yl)pyridine (9.5 g, 34.0 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (1.1 g, 2.7 mmol), and potassium phosphate tribasic monohydrate (23.5 g, 102 mmol) were added to toluene (200 mL) and water (20 mL) and the reaction mixture was degassed with nitrogen. Pd.sub.2(dba).sub.3 (0.622 g, 0.679 mmol) and 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (7.7 mL, 40.8 mmol) were added and the reaction mixture was heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, 12.7 g of amber oil was obtained. The crude material was chromatographed on silica with 9/1 (v/v) hexane/EtOAc to give 7.5 g (77%) of 2-(dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine as a white solid. The product was confirmed by GC/MS and used without further purification.

(88) ##STR00183##

(89) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-isopropylpyridine: 2-(Dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine (7.5 g, 26.3 mmol) was added to a hydrogenator bottle containing EtOH

(150 mL). The reaction mixture was degassed by bubbling nitrogen for 10 min. Pd/C (0.28 g, 2.63 mmol) and Pt/C (0.26 g, 1.3 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane to yield 7.5 g (99%) of the desired product. The product was confirmed by GC/MS and NMR.

(90) ##STR00184##

(91) Synthesis of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine: Dibenzo[b,d]furan-4-ylboronic acid (25 g, 118 mmol), 2,4-dichloropyridine (19.2 g, 130 mmol), Pd(PPh.sub.3).sub.4 (4.1 g, 3.5 mmol) and potassium carbonate (48.9 g, 354 mmol) were added to dimethoxyethane (200 mL) and water (200 mL). The reaction mixture was degassed with nitrogen before being heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane to yield 33.4 g of crude product. The product was crystallized from hexane to give 27.0 g (82%) of 4-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine as white needles. The product was confirmed by GC/MS and NMR.

(92) ##STR00185##

(93) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-(prop-1-en-2-yl)pyridine: 4-Chloro-2-(dibenzo[b,d]furan-4-yl)pyridine (24.0 g, 86.0 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (2.8 g, 6.9 mmol), and potassium phosphate tribasic monohydrate (59.3 g, 257 mmol) were added to toluene (400 mL) and water (40 mL) and the reaction mixture was degassed. Pd.sub.2(dba).sub.3 (1.6 g, 1.7 mmol) and 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (19.4 mL, 103 mmol) were added and the reaction mixture was heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, 33.0 g of amber oil was obtained. The crude material was chromatographed on silica with 9/1 (v/v) DCM/EtOAc to give 23.5 g (96%) of 2-(dibenzo[b,d]furan-4-yl)-4-(prop-1-en-2-yl)pyridine as a white solid. The product was confirmed by GC/MS and used without further purification.

(94) ##STR00186##

(95) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine: 2-(Dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine (8.0 g, 28 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling N.sub.2 for 10 min. Pd/C (0.60 g, 5.6 mmol) and Pt/C (0.55 g, 2.8 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The crude product was chromatographed on silica gel with 9/1 (v/v) hexane/EtOAc to yield 7.2 g (96%) of 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine. The product was confirmed by GC/MS and NMR.

(96) ##STR00187##

(97) Synthesis of 5-bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine: 2,5-Dibromo-4-methylpyridine (30 g, 118 mmol), dibenzo[b,d]furan-4-ylboronic acid (25 g, 118 mmol), Pd(PPh.sub.3).sub.4 (1.4 g, 1.18 mmol), and K.sub.2CO.sub.3 (49 g, 354 mmol) were added to a flask with dimethoxyethane (450 mL) and water (100 mL) and degassed with nitrogen. The reaction mixture was heated to reflux for 15 h before cooling to room temperature. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane and 29.7 g of crude product was obtained. The product was crystallized from hexane to give 28.8 g (72%) of pure product. The product was confirmed by NMR and HPLC (99.3% pure)

(98) ##STR00188##

(99) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine: 5-Bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (28.7 g, 85 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (1.394 g, 3.39 mmol) and potassium phosphate monohydrate (58.6 g, 255 mmol) were added to toluene (500 mL) and water (50 mL) and degassed for 20 min. Trimethylboroxine (14.83 mL, 106 mmol) and Pd.sub.2(dba).sub.3 (0.777 g, 0.849 mmol) were added and the reaction mixture heated to reflux overnight. After cooling, the organic layer was separated and the aqueous layer extracted 3×50 mL with EtOAc, dried over sodium sulfate and evaporated. The crude product was chromatographed on silica gel with 8/2 (v/v) dichloromethane/EtOAc in hexane to give 19.2 g of an off-white solid, which was recrystallized from hexane to give 16.8 g (83%) of the product as white needles. The product was confirmed by NMR and HPLC (99.97% pure).

(100) ##STR00189##

(101) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine: 5-Bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (13.0 g, 38.3 mmol), isobutylboronic acid (11.7 g, 115 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (0.63 g, 1.53 mmol) and potassium phosphate monohydrate (22.1 g, 96 mmol) were mixed in water (10 mL) and toluene (210 mL). The system was degassed for 20 min. with nitrogen and Pd.sub.2(dba).sub.3 (0.35 g, 0.38 mmol) was then added and the system was refluxed overnight. After cooling to room temperature, the reaction mixture was filtered through a small plug of silica gel and eluted with dichloromethane. The filtrate was concentrated and then crystallized from hexane to give 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (9.0 g, 74%).

(102) ##STR00190##

(103) Synthesis of 5-chloro-2-phenylpyridine: 2,5-Dichloropyridine (30 g, 203 mmol), phenylboronic acid (24.72 g, 203 mmol) and potassium carbonate (84 g, 608 mmol) were added to dimethoxyethane (500 mL) and water (100 mL). The reaction mixture was degassed with nitrogen for 20 min, and Pd(PPh.sub.3).sub.4 (2.3 g, 2.0 mmol) was added and reaction mixture was allowed to reflux for 18 h. The reaction was cooled to room temperature, the aqueous layer removed and dimethoxyethane was concentrated to dryness by rotary evaporation under vacuum. The residue was dissolved in DCM and passed through a silica gel plug, eluting with DCM. The solvent was removed and the crude product was chromatographed on silica with 40/60 (v/v) DCM/hexane to 50/50 (v/v) DCM/hexane to yield 28 g (73%) of the product as a white solid (HPLC purity: 99.7%).

(104) ##STR00191##

(105) Synthesis of 5-ethyl-2-phenylpyridine: 5-Chloro-2-phenylpyridine (16 g, 84 mmol) and Ni(dppe)Cl.sub.2 (0.891 g, 1.687 mmol) were added to 300 mL of THF and the reaction mixture was degassed with nitrogen for 20 min. before being cooled to 0° C. Ethylmagnesium bromide (169 mL, 169 mmol) was added dropwise over a period of 60 min. and the reaction mixture stirred for an additional 3 h at before warming to room temperature overnight. The reaction mixture was recooled to 0° C. and quenched with 250 mL of water, extracted with EtOAc and the organic layer dried over sodium sulfate and filtered. The crude material was chromatographed on silica with 95/5 hexane/EtOAc to give 2.9 g (19%) of 5-ethyl-2-phenylpyridine as a white solid.

(106) ##STR00192##

(107) Synthesis of 2-phenyl-5-(prop-1-en-2-yl)pyridine: To a 1 L round bottom flask was added 5-chloro-2-phenylpyridine (10.15 g, 53.5 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (1.8 g, 4.3 mmol), potassium phosphate tribasic monohydrate (37.0 g, 161 mmol) with toluene (200 mL) and water (20 mL). The reaction mixture was degassed with nitrogen for 20 mins. 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (12.07 mL, 64.2 mmol) and Pd.sub.2(dba).sub.3 (0.980 g, 1.070 mmol) were added and the reaction mixture was refluxed for 18 h. The aqueous layer was removed and the organic layer was concentrated to dryness. The crude product was chromatographed on silica gel with 0-20% EtOAc in hexane to yield 11 g of the desired product (HPLC purity: 95%). The product was confirmed by GC/MS.

(108) ##STR00193##

(109) Synthesis of 2-phenyl-5-isopropylpyridine: 2-Phenyl-5-(prop-1-en-2-yl)pyridine (11 g, 56.3 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling N.sub.2 for 10 min. Pd/C (0.60 g, 5.63 mmol) and Pt/C (0.55 g, 2.82 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1.5 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The solvent was removed on rotoevaporator and GC/MS confirmed complete conversion. The crude product was adsorbed on Celite® for column chromatography. The crude product was chromatographed on silica gel with 10% EtOAc in hexane to yield 6 g (54%) of 2-phenyl-5-isopropylpyridine (HPLC purity: 100%). The product was confirmed by GC/MS.

(110) ##STR00194##

(111) Synthesis of 4-chloro-2-phenylpyridine: A 1 L round bottom flask was charged with 2,4-dichloropyridine (30 g, 203 mmol), phenylboronic acid (24.7 g, 203 mmol), potassium carbonate (84 g, 608 mmol), Pd(PPh.sub.3).sub.4 (2.3 g, 2.0 mmol), dimethoxyethane (500 mL) and water (150 mL). The mixture was degassed and heated to reflux for 20 h. After cooling, the aqueous layer was extracted with EtOAc; the organic portion was combined and subjected to column chromatography (SiO.sub.2, 5% EtOAc in hexane to 10% EtOAc in hexane) to give 34 g (88%) of 4-chloro-2-phenylpyridine. The product was confirmed by GC/MS and NMR.

(112) ##STR00195##

(113) Synthesis of 2-phenyl-4-(prop-1-en-2-yl)pyridine: 4-Chloro-2-phenylpyridine (14 g, 73.8 mmol) and potassium phosphate (51.0 g, 221 mmol) were dissolved in 300 mL of toluene and 30 mL of water. The reaction was purged with nitrogen for 20 minutes and then 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (16.65 mL, 89 mmol), Pd.sub.2(dba).sub.3 (1.35 g, 1.48 mmol) and S-Phos (2.42 g, 5.91 mmol) were added. The reaction was brought to reflux for 18 h. After cooling, 100 mL of water was added, separated and the aqueous layer extracted twice with 100 mL of ethyl acetate. The organic layers were passed through a plug of silica gel, eluting with DCM. After evaporation of the solvent, the crude product was subjected to column chromatography (SiO.sub.2, 5% EtOAc in hexane to 10% EtOAc in hexane) to get 13.5 g (90%) of 2-phenyl-4-(prop-1-en-2-yl)pyridine.

(114) ##STR00196##

(115) Synthesis of 2-phenyl-4-isopropylpyridine: 2-Phenyl-4-(prop-1-en-2-yl) pyridine (13.5 g, 69.1 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling with nitrogen for 10 min. Pd/C (0.736 g, 6.9 mmol) and Pt/C (0.674 g, 3.5 mmol) was added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 2 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The solvent was removed on a rotary evaporator and GC/MS confirmed complete conversion. The crude product was adsorbed on Celite® for column chromatography. The crude product was chromatographed on silica gel with 10% EtOAc in hexane to yield 10 g (75%) of 2-phenyl-4-isopropylpyridine (HPLC purity: 99.8%). The product was confirmed by GC/MS.

(116) ##STR00197##

(117) Synthesis of 5-methyl-2-phenylpyridine: 2-Bromo-5-methylpyridine (30 g, 174 mmol), phenylboronic acid (25.5 g, 209 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (2.86 g, 6.98 mmol) and potassium phosphate tribasic monohydrate (120 g, 523 mmol) were added to toluene (600 mL) and water (60 mL). The reaction mixture was degassed with nitrogen for 20 min. Pd.sub.2(dba).sub.3 (3.19 g, 3.49 mmol) was added and the reaction mixture was refluxed for 18 h. After cooling, the organic layer was separated and the aqueous layer extracted with 3×50 mL dichloromethane, dried over sodium sulfate and evaporated. The crude product was chromatographed on silica gel with 75/25 (v/v) hexane/EtOAc and then distilled on a Kugelrohr apparatus (150° C., 100 mbar) to give 26 g (88%) of 5-methyl-2-phenylpyridine as a

white solid. The product was confirmed by NMR and GC/MS. HPLC purity: 99.2%.

(118) ##STR00198##

(119) Synthesis of 4-methyl-2-phenylpyridine: A 1 L round bottom flask was charged with 2-chloro-4-methylpyridine (25 g, 196 mmol), phenylboronic acid (23.9 g, 196 mmol), potassium carbonate (81 g, 588 mmol), Pd(PPh₃)₄ (2.3 g, 1.9 mmol), dimethoxyethane (500 mL) and water (150 mL). The reaction mixture was degassed with nitrogen and heated to reflux for 22 h. After cooling, the aqueous layer was extracted with EtOAc; the organic portion was combined and subjected to column chromatography (SiO₂, 5% EtOAc in hexane to 10% EtOAc in hexane) to give 28 g (78%) of 4-methyl-2-phenylpyridine. The product was confirmed by NMR and GC/MS.

(120) ##STR00199##

(121) Synthesis of 4-ethyl-2-phenylpyridine: To 4-methyl-2-phenylpyridine (8 g, 47.3 mmol) in dry THF (150 mL) at -78° C. was added dropwise lithium diisopropylamide (LDA) (30.7 mL, 61.5 mmol). The dark solution was stirred for 3 h at -78° C. and then CH₃I was added (4.1 mL, 66.2 mmol) dropwise. The reaction mixture was allowed to slowly warm to room temperature overnight. Ammonium chloride solution and EtOAc were added and the reaction transferred to a separatory funnel. The layers were separated, washing the aqueous twice with EtOAc and combined organics once with water. After removal of the solvent, the crude product was chromatographed on silica gel with 9/1 (v/v) hexane/EtOAc to give 5.5 g (63.5%) of 4-ethyl-2-phenylpyridine. HPLC purity: 99.0%.

(122) ##STR00200##

(123) Synthesis of 4-methyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-methyl-2-phenylpyridine (7 g, 41 mmol) and iridium(III) chloride hydrate (4.86 g, 13.79 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 7.5 g (90%) of the desired product. The product was used without further purification.

(124) ##STR00201##

(125) Synthesis of 5-methyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round bottom flask was added 5-methyl-2-phenylpyridine (12 g, 70.9 mmol) and iridium(III) chloride hydrate (7.1 g, 20.3 mmol) with 2-ethoxyethanol (100 mL) and water (33.3 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 11.0 g (96%) of the desired product. The product was used without further purification.

(126) ##STR00202##

(127) Synthesis of 2-phenyl-5-isopropylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 5-isopropyl-2-phenylpyridine (6.0 g, 30.4 mmol) and iridium(III) chloride hydrate (3.6 g, 10.1 mmol) with 2-ethoxyethanol (100 mL) and water (33.3 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 7 g (100%) of the desired product. The product was used without further purification.

(128) ##STR00203##

(129) Synthesis of 2-phenyl-4-isopropylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-isopropyl-2-phenylpyridine (8.0 g, 40.6 mmol) and iridium(III) chloride hydrate (7.4 g, 20.3 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 6.1 g (95%) of the desired product. The product was used without

further purification.

(130) ##STR00204##

(131) Synthesis of 4-ethyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-isopropyl-2-phenylpyridine (5.5 g, 30.0 mmol) and iridium(III) chloride hydrate (5.8 g, 16.5 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 6.5 g (72%) of the desired product. The product was used without further purification.

(132) ##STR00205##

(133) Synthesis of 5-ethyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round bottom flask was added 5-ethyl-2-phenylpyridine (2.9 g, 15.7 mmol) and iridium(III) chloride hydrate (1.8 g, 5.2 mmol) with 2-ethoxyethanol (60 mL) and water (20 mL) under a nitrogen atmosphere. The reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 2.45 g (89.3%) of the desired product. The product was used without further purification.

(134) ##STR00206##

(135) Synthesis of 5-methyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (11 g, 9.8 mmol) was suspended in 600 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (5.3 g, 20.5 mmol) was dissolved in MeOH (300 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 15 g (100%) of product as a brownish green solid. The product was used without further purification.

(136) ##STR00207##

(137) Synthesis of 4-methyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (7.5 g, 6.6 mmol) was dissolved in 600 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (3.5 g, 13.8 mmol) was dissolved in MeOH (300 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 10 g (100%) of product as a brownish green solid. The product was used without further purification.

(138) ##STR00208##

(139) Synthesis of 2-phenyl-5-isopropylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (5.3 g, 4.3 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (2.3 g, 8.9 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 6.9 g (100%) of product as a brownish solid. The product was used without further purification.

(140) ##STR00209##

(141) Synthesis of 2-phenyl-4-isopropylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (6.2 g, 4.94 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (2.7 g, 10.4 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 7.8 g (100%) of product as a brownish green solid. The product was used without further purification.

(142) ##STR00210##

(143) Synthesis of 4-ethyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (6.8 g, 5.7 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I)

trifluoromethanesulfonate (3.2 g, 12.5 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 5.5 g (63%) of product as a brownish green solid. The product was used without further purification.

(144) ##STR00211##

(145) Synthesis of 5-ethyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (2.8 g, 2.4 mmol) was suspended in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (1.3 g, 4.91 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 3.6 g (100%) of product as a brownish green solid. The product was used without further purification.

(146) ##STR00212##

(147) Synthesis of Compound 53: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.5 g, 4.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isopropylpyridine (3.5 g, 12.18 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.3 g (33%) of Compound 53 as a yellow solid. The product was confirmed by HPLC (99.5% pure) and LC/MS.

(148) ##STR00213##

(149) Synthesis of Compound 157—A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.50 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was refluxed for 24 h. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane to yield a mixture of fac- and mer-isomers by LC/MS. The mixture was isomerized to the fac-isomer in a Rayonet at 350 nm in DMSO. The crude product was chromatographed on silica gel with 1/1 dichloromethane/hexane to yield 1.4 g (52%) of Compound 157 as a yellow solid. The product was confirmed by HPLC (98.7% pure) and LC/MS.

(150) ##STR00214##

(151) Synthesis of Compound 158: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 2.7 g (100%) of Compound 158 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

(152) ##STR00215##

(153) Synthesis of Compound 159: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.04 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.0 g, 10.98 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solubility of the desired product was very poor. Large amount of solvent was used to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane and later 4/1 dichloromethane/hexane to yield 0.3 g of the product as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

(154) ##STR00216##

(155) Synthesis of Compound 165: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.2 g (43%) of Compound 165 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

(156) ##STR00217##

(157) Synthesis of Compound 174: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.6 g, 4.68 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.6 g, 13.17 mmol) in EtOH (50 mL) and MeOH (50 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 0.8 g of product as a yellow solid was confirmed by HPLC (98.6% pure) and LC/MS.

(158) ##STR00218##

(159) Synthesis of Compound 175: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.66 g, 9.74 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) THF/hexane to yield 0.8 g of product by HPLC. The product was recrystallized by slow evaporation of DCM from a 1/3 DCM/hexane solution to yield 0.6 g (22%) as a yellow crystalline solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

(160) ##STR00219##

(161) Synthesis of Compound 184: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 3.76 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.0 g, 10.98

mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.1 g (65%) of product as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.

(162) ##STR00220##

(163) Synthesis of Compound 185: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.8 g, 3.51 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethyl pyridine (2.88.0 g, 10.53 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under N.sub.2 atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 2.1 g (69%) of product as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

(164) ##STR00221##

(165) Synthesis of Compound 314: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (2.5 g, 7.93 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 3.0 g (100%) of Compound 314 as a yellow solid. The product was confirmed by HPLC (99.6% pure) and LC/MS.

(166) ##STR00222##

(167) Synthesis of Compound 321: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.2 g, 2.86 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (2.2 g, 6.98 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.6 g (50%) of Compound 321 as a yellow solid. The product was confirmed by HPLC (99.0% pure) and LC/MS.

(168) ##STR00223##

(169) Synthesis of Compound 625: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.2 g, 3.08 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.2 g, 7.66 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the

volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 1.7 g (67%) of Compound 625 as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.

(170) ##STR00224##

(171) Synthesis of Compound 626: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.5 g, 8.70 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.5 g (89%) of Compound 626 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

(172) ##STR00225##

(173) Synthesis of Compound 627: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.0 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 10.4 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.0 g (60%) of Compound 627 as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

(174) ##STR00226##

(175) Synthesis of Compound 628: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.0 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 10.5 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 24 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 2.1 g (64%) of product as a yellow solid. The product was confirmed by HPLC (99.95% pure) and LC/MS.

(176) ##STR00227##

(177) Synthesis of Compound 633: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.5 g, 8.70 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.6 g (59%) of Compound 633 as a yellow solid. The product was confirmed by HPLC (99.7% pure) and LC/MS.

(178) ##STR00228##

(179) Synthesis of Compound 643: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.4 g, 3.12 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-ethylpyridine (2.69 g, 9.35 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 1.3 g (50%) of product as a yellow solid. The product was confirmed by HPLC (100% pure) and LC/MS.

(180) ##STR00229##

(181) Synthesis of Compound 652: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.1 g, 3.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.1 g, 10.9 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.1 g (62%) of Compound 652 as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS

(182) ##STR00230##

(183) Synthesis of Compound 653: A mixture of iridium trifluoromethanesulfonate complex (2.4 g, 3.01 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 9.02 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 0.96 g (45%) of product as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.

(184) ##STR00231##

(185) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-ethyl-*d.sub.3*-pyridine: To 2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (15.3 g, 59.0 mmol) in dry THE (250 mL) at -78° C. was added dropwise lithium diisopropylamide (35.4 mL, 70.8 mmol). The dark solution was stirred for 2 h at -78° C. and then CD.*sub.3*I was added (4.41 mL, 70.8 mmol) dropwise. The reaction mixture was allowed to slowly warm to RT overnight. Ammonium chloride solution and EtOAc were added and the reaction transferred separatory funnel. The layers were separated, washing the aqueous twice with EtOAc and combined organics once with water. After removal of the solvent, the crude product was chromatographed on silica gel with 8/2 (v/v) hexane/EtOAc and then 7/3 hexane/EtOAc to give 14.5 g of product as a pale yellow solid. Recrystallization from hexane gave 12.9 g (79%) of 2-(dibenzo[b,d]furan-4-yl)-4-ethyl-*d.sub.3*-pyridine. HPLC purity: 99.4%.

(186) ##STR00232##

(187) Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-isopropyl-*d.sub.6*-pyridine: 2-(Dibenzo[b,d]furan-4-yl)-4-ethyl-*d.sub.3*-pyridine was dissolved in dry THE (100 mL) and cooled to -78° C. Lithium diisopropylamide (19.0 mL, 38.0 mmol) was added dropwise and the reaction mixture stirred for 2 h at -78° C. CD.*sub.3*I was added dropwise and the reaction mixture was allowed to slowly warm to room temperature overnight. The reaction was quenched with MeOH, NH.*sub.4*Cl (aq.) and EtOAc were added and the biphasic mixture transferred to a separatory funnel. The layers were separated, washing the aqueous layer twice with EtOAc and the combined organics with water. After removal of the solvent, the crude product was chromatographed on silica gel with 8/2 (v/v)

hexane/EtOAc to give 6.4 g (86%) of 2-(dibenzo[b,d]furan-4-yl)-4-isopropyl-d.sub.6-pyridine.
HPLC purity: 99.2%.

(188) ##STR00233##

(189) Synthesis of Compound 1145: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.5 g, 4.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-d.sup.3-ethylpyridine (3.5 g, 12.7 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, celite was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with dichloromethane to yield 1.8 g (47%) of Compound 1145 as a yellow solid. The product was confirmed by HPLC (98.7% pure) and LC/MS.

(190) ##STR00234##

(191) Synthesis of Compound 1146: 2-(Dibenzo[b,d]furan-4-yl)-4-isopropyl-d.sub.6-pyridine and the appropriate iridium trifluoromethanesulfonate complex were combined in ethanol (25 mL) and methanol (25 mL) and heated to reflux for 16 h. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The celite/silica plug was then washed with dichloromethane to dissolve the product. The crude product was chromatographed on silica gel with 50-70% dichloromethane in hexane to and then sublimed to yield 1.7 g (43%) of Compound 1146 as a yellow solid. The product was confirmed by HPLC (99.5% pure) and LC/MS.

(192) It is understood that the various embodiments described herein are by way of example only, and are not intended to limit the scope of the invention. For example, many of the materials and structures described herein may be substituted with other materials and structures without deviating from the spirit of the invention. The present invention as claimed may therefore include variations from the particular examples and preferred embodiments described herein, as will be apparent to one of skill in the art. It is understood that various theories as to why the invention works are not intended to be limiting.

Claims

1. A compound having the formula: ##STR00235## wherein R.sub.1 and R.sub.2 are optionally linked; wherein the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2; wherein R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked; wherein R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution; wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO.sub.2, SiRR', and GeRR'; wherein R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2; wherein at least one R.sub.a is nitrile; and wherein at least one of R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.

2. The compound of claim 1, wherein n is 2.

3. The compound of claim 1, wherein X is O.

4. The compound of claim 1, wherein R.sub.1 is hydrogen and R.sub.2 is selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

5. The compound of claim 1, wherein R.sub.1 is selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof and R.sub.2 is hydrogen.
6. The compound of claim 1, wherein R.sub.1 and R.sub.2 are selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.
7. The compound of claim 1, wherein R.sub.1 or R.sub.2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.
8. The compound of claim 1, wherein R.sub.1 or R.sub.2 contain one or more deuterium atoms.
9. The compound of claim 1, wherein R.sub.1 and R.sub.2 contain one or more deuterium atoms.
10. The compound of claim 1, wherein R.sub.3, R.sub.4, R.sub.5 and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof.
11. The compound of claim 1, wherein at least one of R.sub.3, R.sub.4, R.sub.5 and R.sub.6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl.
12. The compound of claim 1, wherein R.sub.3, R.sub.4, R.sub.5 or R.sub.6 contain one or more deuterium atoms.
13. The compound of claim 1, wherein R.sub.4 and R.sub.5 are selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.
14. The compound of claim 1, wherein R.sub.4 and R.sub.5 are selected from the group consisting of halide, alkyl, cycloalkyl, silyl, aryl, heteroaryl, nitrile, and combinations thereof.
15. The compound of claim 1, wherein R.sub.4 is alkyl.
16. The compound of claim 1, wherein R.sub.5 is alkyl.
17. The compound of claim 1, wherein R.sub.4 is silyl.
18. The compound of claim 1, wherein R.sub.5 is silyl.
19. A first device comprising a first organic light emitting device, comprising: an anode; a cathode; and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula: ##STR00236## wherein R.sub.1 and R.sub.2 are optionally linked; wherein the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2; wherein R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked; wherein R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution; wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO.sub.2, SiRR', and GeRR'; wherein R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2; wherein at least one R.sub.a is nitrile; and wherein at least one of R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.
20. A consumer product comprising a first organic light emitting device, comprising: an anode; a cathode; and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula: ##STR00237## wherein R.sub.1 and R.sub.2 are optionally linked; wherein the sum of the number of carbon atoms in R.sub.1 and R.sub.2 is at least 2; wherein R.sub.3, R.sub.4, R.sub.5, R.sub.6 are optionally linked; wherein R.sub.a and R.sub.b represent mono-, di-, tri- or tetra-substitution; wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO.sub.2, SiRR', and GeRR'; wherein R.sub.a, R.sub.b, R, R', R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 are independently selected from the group consisting of

hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2; wherein at least one R.sub.a is nitrile; and wherein at least one of R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, and R.sub.6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.
